

EDA 05: Reviews Table — EDA and Data Quality

1. Dataset Overview

- 99,224 rows × 7 columns: Review ID, order ID, review score, comment title, comment message, creation date, and answer timestamp.
- Review scores range from 1 to 5. Comment fields contain free text in Brazilian Portuguese.
- Review dates range from 2016-10-02 to 2018-08-31, while answers extend to 2018-10-29. Orders span 2016-09-04 to 2018-10-17.

2. Duplicates and Multiple Reviews

- No, fully duplicated rows; (review_id, order_id) is unique.

3. Missing Values

- Review title missing: 87,656 (88.34%); review message missing: 58,247 (58.70%).
- 27 messages contain only spaces.
- 56,536 reviews have neither title nor message.
- Decision: Treat blank comments as missing and do not impute review text.

4. Review Score Distribution

- 5 stars: 57.8%, 4 stars: 19.3%, 3 stars: 8.2%, 2 stars: 3.2%, 1 star: 11.5%.
- The distribution is U-shaped, with most reviews receiving 5 or 1 star. No invalid scores were found.

5. Missing Comments and Sentiment

- Messages are present in 76.5% of 1-star reviews versus 35.8% of 5-star reviews. Median message lengths are 89 and 42 characters, respectively.
- Common positive phrases include muito bom, bom, and recomendo.
- nao recebi (not received) appears in 16.7% of bad-review comments versus 0.3% of good-review comments. Other complaints include delays, defects, wrong items.

6. Review Response Time

- Median response time: 1.67 days; 95th percentile: 6.97 days; 99th percentile: 22.07 days.
- 682 responses took over 30 days, including 172 over 100 days. Maximum: 518.7 days.
- No responses occurred before review creation. Median response times are similar across scores (1.55–1.79 days).
- The only extreme values are late answers: 172 after more than 100 days, up to 518.7 days. The answer date looks valid, so the customer simply answered very late.

EDA 06: Products Table — EDA and Data Quality

1. Dataset Overview

- 32,951 rows × 9 columns: Product ID, category, listing attributes (name length, description length, photos), weight, and three dimensions.
- Two source column names are misspelled: `product_name_lenght` and `product_description_lenght`.
- Decision: Rename them to `product_name_length` and `product_description_length` in the working dataset.

2. Missing Values

- 610 products (1.85%) have missing category, name length, description length, and photo count, indicating incomplete product listings.
- Decision: Retain these products and handle missing values appropriately during preprocessing rather than dropping them.

3. Invalid and Extreme Values

- Four products have 0 g weight, which is physically implausible. All belong to `cama_mesa_banho` and share dimensions of 30 × 25 × 30 cm.
- The heaviest product weighs 40,425 g but has dimensions of only 13 × 65 × 28 cm, suggesting a potential data inconsistency (Explained in last section).
- Decision: Treat zero weight as unknown. Retain extreme values initially and apply capping or other robust handling within the model pipeline, if justified by the feature distributions.

4. Feature Relationships

- Product weight and box volume have a correlation of 0.80, indicating a strong positive relationship, which is reasonable for physical products.
- Decision: Retain both features, while checking for outliers and potential redundancy during feature engineering.

5. Category Translation

- Seven category names are identical in Portuguese and English, including `audio`, `cool_stuff`, `pet_shop`, and `market_place`.
- Decision: Keep these mappings unchanged because the terms are already used in English; identical translations do not necessarily indicate errors.

Why this is suspicious:

- Weight: 40,425 g (about 40 kg).
- Box size: 13 × 65 × 28 cm, with a volume of about 24 litres.
- A 40 kg product in such a small box is unusual for common marketplace products like clothes, toys, or household items.
- The weight may have been entered incorrectly, possibly due to an extra digit. For example, 4,042 g (about 4 kg) would be more reasonable for this box size.
- However, the product could be a heavy item, such as metal parts or tools, or the dimensions could be incorrect.

EDA Part 7: Seller Data — EDA and Data Quality

1. Dataset Overview

- 3,095 rows × 4 columns: Seller ID, ZIP code prefix, city, and state.
- Each row represents one seller. Product details and seller performance must be derived by joining other tables.
- seller_id is unique, with no duplicate rows or missing values.

2. Geographic Distribution

- Sellers operate across 23 states.
- São Paulo (SP) accounts for 1,849 sellers (59.7%); the top five states (SP, PR, MG, SC, RJ) account for 90.6%.
- Five states (AC, PI, MA, AM, PA) have only one seller each.
- There are 611 distinct city names. Top cities include São Paulo (694 sellers), Curitiba (127), Rio de Janeiro (96), Belo Horizonte (68), and Ribeirão Preto (52).
- 342 cities (56.0%) have only one seller, indicating a long-tailed distribution.

3. City Name Quality

- City names are mostly lowercase with no extra spaces. Three values contain accents; removing accents reduces distinct names from 611 to 609.
- One city value (04482255) appears to be a ZIP code mistakenly entered as a city.
- Twenty-five city values contain symbols or additional location text, e.g., sao paulo / sao paulo and novo hamburgo, rio grande do sul, brasil.
- Thirty-four ZIP prefixes have multiple city-name variants, including spelling errors such as garulhos / guarulhos and sando andre / santo andre.

Decision: Do not use city names as a reliable join key. Avoid manual corrections based on assumptions; use ZIP prefixes for location mapping.

4. ZIP Prefix and State Inconsistencies

- 17 ZIP prefixes (45 sellers) are associated with multiple states, suggesting incorrect state assignments.
- Examples:
 - ZIP prefix 21210: Rio de Janeiro city mapped to both RJ and RN.
 - ZIP prefix 22783: Rio de Janeiro city mapped to both RJ and SP.
 - ZIP prefix 31160: Belo Horizonte city mapped to both MG and SP.
- Twenty-three city names appear across multiple states, partly reflecting these inconsistencies.
- Customers span 27 states, with approximately 42% in SP, indicating potential cross-state orders.

Decision: Retain all seller records because there are no missing or duplicate rows. Treat seller_state as potentially unreliable and cross-check it against the cleaned geolocation lookup after joining on ZIP prefix.

EDA Part 8: Geolocation Data — EDA and Data Quality

1. Dataset Overview

- 1,000,163 rows × 5 columns: ZIP code prefix, latitude, longitude, city, and state.
- Largest table, containing approximately 10× more rows than the orders table.
- Each row represents a coordinate point, not a unique ZIP code prefix. Multiple points can exist for the same prefix.

2. Duplicate Analysis

- Fully duplicated rows: 261,831 (26.2%).
- Rows remaining after exact deduplication: 738,332.
- Repeated (prefix, latitude, longitude) combinations: 280,009.
- Distinct (prefix, latitude, longitude) combinations: 720,154.

3. Missing Values and Geographic Validity

- No missing values were found in any column.
- Latitude ranges from -36.61 to 45.07 and longitude from -101.47 to 121.11, exceeding Brazil's geographic boundaries.
- Only 33 rows (0.004%), across 21 prefixes, fall outside the expected Brazil bounding box.
- Decision: Exclude points outside Brazil from the geolocation lookup because they are geographically invalid and account for a negligible share of the data.

4. City Name Quality

- Non-ASCII city text: 63,436 rows (8.6% after exact deduplication).
- Distinct city names: 8,011 raw, reduced to 5,967 after accent removal.
- São Paulo appears under two spellings: sao paulo (79,927 rows) and são paulo (19,718 rows).
- Nineteen city names contain symbols, digits, or brackets; two contain web-encoded characters.
- 8,556 prefixes (45.0%) have more than one city name, indicating inconsistent naming.
- Decision: Ignore city names as a reliable join key because of spelling, accent, and encoding inconsistencies.

5. State Consistency

- Eight ZIP code prefixes appear under multiple states, indicating incorrect state assignments.
- No invalid state codes were identified.
- Decision: Assign each prefix its most frequently observed state and use this as the state lookup.